2023 Princeton University Science Olympiad Invitational
Dynamic Planet: Fresh Water

TEST BOOKLET

Instructions

Do not open the test booklet until you are instructed to do so, but you may write your name, school, and team number on the test booklet and the answer sheet.

For multiple choice, matching, and true/false, write all answers on the answer sheet provided. Answer short answer questions in the test booklet.

You have 50 minutes to complete this test.

Materials

You are allowed the following materials:

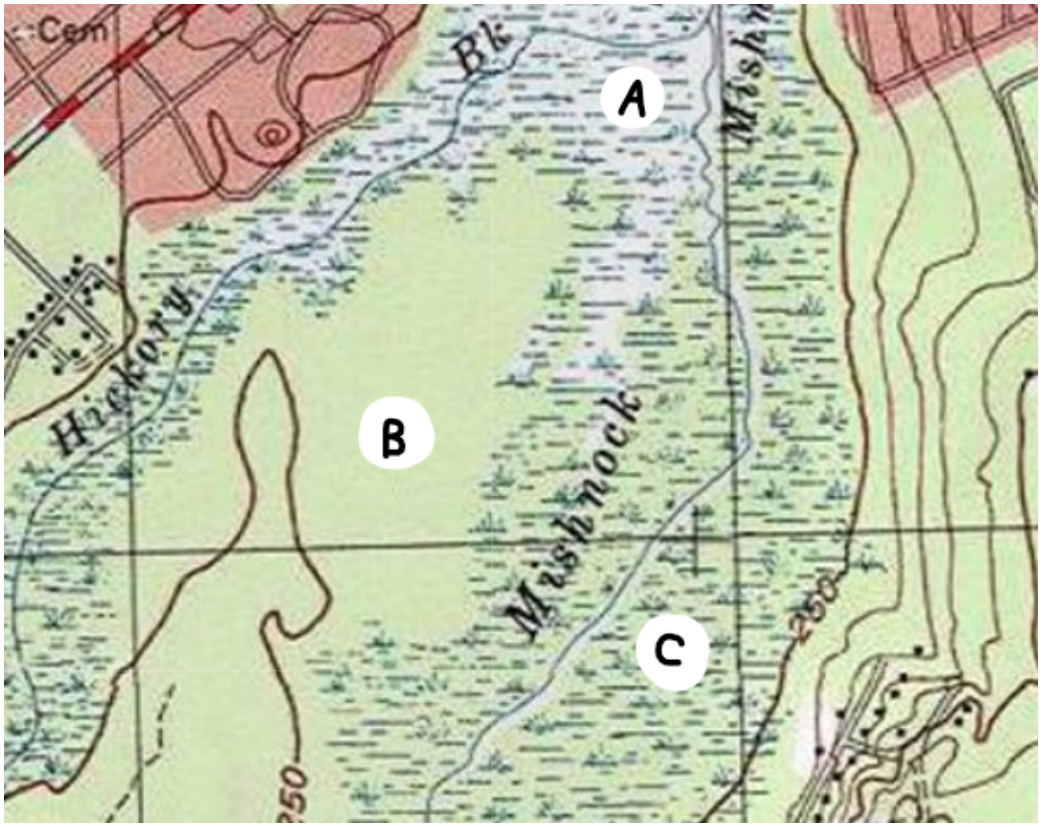
- one three-ring binder of any size containing information in any form and from any source attached using the available rings. Sheet protectors, lamination, tabs and labels are permitted.
- two stand-alone calculators of any type.
- writing instrument + erasers

Test Structure

The test contains a mixture of multiple choice, matching, true/false, and short answer questions. Information about the test structure is provided below.
Test Total: 220 pts. 99 questions.
Multiple Choice: 120 pts, 2 pts each question. 60 questions.
Matching & True/False: 17 pts, 1 pt each question. 17 questions.
Short Answer: 83 pts, point value varies by question. 22 questions.

The indicated tiebreaker questions will be used to break ties.

Multiple Choice (120 pts total, 2 pts each question)
Write your answers on the provided answer sheet.



The map above represents a region in the Pawtuxet River Valley of Rhode Island. Use the map to answer question 1.

1. Which of the following statements are **true**?
   - I. The land at A is a marsh or swamp
   - II. The land at C is submerged year-round
   - III. The lands at A and C differ because C is submerged year-round and A is not

   a. Only I is true
   b. Only II is true
   c. Only I and II are true
   d. All of the statements are true

2. When do dendritic drainage patterns form?
   a. When there are faults in the rock
   b. When the rock underneath the land is uniformly resistant to erosion
   c. When the elevation rapidly drops from high to low
   d. When riparian buffers are scattered in the rock

3. Continental divides separate:
    a. Major reservoirs
    b. River valleys
    c. Continent landmasses
    d. Major drainage basins

4. Where would a continental divide most likely form?
    a. At the thalweg of a valley
    b. Along a young stream
    c. Along a dormant tectonic plate edge
    d. Along recently formed mountain ranges

5. Which of the following statements about ephemeral streams is **correct**?
    a. Groundwater is the primary source of water for ephemeral streams
    b. There is flowing water in ephemeral streams for most of the year, but not always
    c. The water table is generally not located above the streambed for ephemeral streams
    d. Ephemeral streams are also known as intermittent streams

6. Where do ephemeral streams tend to occur?
    a. In temperate, moist climates
    b. In dry climates with low rainfall
    c. In salt-heavy benthic zones
    d. In dry climates with low evaporation rates

7. Which of the following factors would **not** significantly affect stream velocity?
    a. The position within the channel (i.e. the sides vs. the middle)
    b. The quantity of rocks, pebbles, and boulders embedded in the stream channel
    c. The acidity of the stream water
    d. The gradient of the stream

8. What is the primary difference between laminar and turbulent flows?
    a. In laminar flow, particles travel in similar parallel paths, whereas in turbulent flow, particles travel in irregular paths
    b. In laminar flow, particles travel in irregular paths, whereas in turbulent flow, particles travel in similar parallel paths
    c. In laminar flow, particles are dense and erode stream channels more quickly, whereas in turbulent flow, particles are less dense and erode stream channels very slowly
    d. In laminar flow, particles are less dense and erode stream channels very slowly, whereas in turbulent flow, particles are dense and erode stream channels more quickly

9. Which of the following statements are **true**?

I. Streams on the surface generally have a higher dissolved load than those with sources with a deep underground source

II. High velocity currents can carry larger and denser particles than low velocity currents

III. Sediment can change in between bed load and suspended load as stream velocity changes

   a. Only II is true
   b. Only III is true
   c. I and II are true
   d. II and III are true

10. Competence and capacity ____ with increasing discharge.
    a. Increase
    b. Decrease
    c. Remain the same

11. Moving downstream, which of the following is true?
    a. Discharge decreases as water leaves the stream through tributary streams and groundwater
    b. The width, depth, and average velocity of the stream decrease
    c. The size of particles making up bed load tends to decrease
    d. Velocity decreases

12. When a dam impedes stream flow,
    a. The long profile remains the same, but erosion takes place downstream
    b. The gradient increases as the rate of sediment deposition decreases
    c. Erosion takes place downstream of the dam, and the stream adjusts to a new local base level
    d. The dam adjusts to match the regional base level

13. What is a primary difference between a valley and a canyon? **(VOIDED)**
    a. Valleys have gently sloping sidewalls while canyons have steep sidewalls that form cliffs
    b. Valleys are predominantly formed by lateral erosion while canyons are predominantly formed by downcutting
    c. Streams only run through valleys, not canyons
    d. Weak rocks yield the vertical cliffs of canyons, and strong rocks produce the gently sloped canyon walls

14. The Niagara Falls is estimated to be receding southward to Lake Erie at a rate of 1 foot per year. What is the most plausible explanation for this phenomenon?
    a. Gravitational forces warp the water's directional velocity, forcing the waterfall southward
    b. Active glaciers underneath are moving the bedrock to the south
    c. Erosion occurs at the top of the waterfall, pushing the overhang southward
    d. Erosion occurs at the base of the waterfall, depressing the rock southward

For questions 21-24, decide if the stream described is a headwater stream, medium stream, or river. Stream orders are given by the Strahler number.

15. This stream is located in the upper reaches of its watershed, generally forming on steep slopes.
    a. Headwater stream
    b. Medium stream
    c. River

16. This stream represents over 80% of the world's waterways and can be any stream order from first- through third- order.
    a. Headwater stream
    b. Medium stream
    c. River
17. This stream is classified as a twelfth-order waterway.
    a. Headwater stream
    b. Medium stream
    c. River

18. This stream is relatively not steep and flows more slowly, with high amounts of debris and runoff. Its order is at least seventh order.
    a. Headwater stream
    b. Medium stream
    c. River

19. What is the point where a tributary meets the mainstem?
    a. affluence
    b. fork
    c. deferred junction
    d. confluence

20. What are the streams that result from river bifurcation called?
    a. Yazoo tributaries
    b. distributaries
    c. Rosgen streams
    d. riffles

21. Which of the following statements about meanders are **true**?
    I. Meanders are most often formed in alluvial materials.
    II. A gooseneck is an extremely bowed meander that has been sealed off from the stream.
    III. The erosional energy on the outer banks of meanders tends to concentrate in the middle of the channel.

    a. Only I is true
    b. Only II is true
    c. Only III is true
    d. I and III are true

22. What is a primary difference between floodplains and wetlands?
    a. Wetlands encompass floodplains, which always form within existing wetlands
    b. Floodplains mostly pose a threat to nearby communities, whereas wetlands are beneficial
    c. Floodplains only periodically flood whereas wetlands are generally consistently saturated
    d. The terms 'floodplain' and 'wetland' are interchangeable

For questions 29-33, decide if the stream channel depicted is braided, meandering, straight, or anastomosing.

23.



    a. Braided
    b. Meandering
    c. Straight
    d. Anastomosing

24.



a. Braided
b. Meandering
c. Straight
d. Anastomosing

25.



a. Braided
b. Meandering
c. Straight
d. Anastomosing

26.



a. Braided
b. Meandering
c. Straight
d. Anastomosing

27.



a. Braided
b. Meandering
c. Straight
d. Anastomosing

28. In which of the following streams would sinuosity tend to be highest?
    a. Stream channels with little riparian vegetation
    b. Stream channels on steep gradients
    c. Stream channels with high resistance
    d. None of the above

29. By what aspect of the individual particles are sediments sorted during transportation?
    a. Weight
    b. Caliche
    c. Density
    d. Uniformity

30. In which of the following scenarios would stream competence increase?
    a. Eddying currents decrease in strength
    b. Stream velocity increases
    c. Discharge decreases
    d. Discharge increases

31. How do aquifuges, aquitards, and aquicludes differ in terms of permeability?
    a. Aquitards are areas in aquifers with low permeability; aquicludes and aquifuges are completely impermeable
    b. Aquitards are areas in aquifers that are completely impermeable, aquicludes and aquifuges have low permeability
    c. Aquicludes are areas in aquifers with low permeability; aquitards and aquifuges are completely impermeable
    d. Aquicludes are areas in aquifers that are completely impermeable, aquitards and aquifuges have low permeability

32. Holding all other factors constant, changing which of the following factors would not directly impact the storativity of an aquifer?
    a. Hydraulic head
    b. Thickness of the aquifer
    c. Specific yield of the aquifer
    d. Elevation of the aquifer

33. **(TB2)** Which of the following statements about aquifers are true?
    I. In isotropic aquifers, hydraulic conductivity is equal in all directions.
    II. Anisotropy typically occurs in semi-confined aquifers.
    III. Karst aquifers are typically highly anisotropic.

    a. Only I is true
    b. Only II is true
    c. II and III are true
    d. All of the statements are true

34. Why does saltwater remain below freshwater in aquifers?
    a. Saltwater intrusion displaces the freshwater column
    b. Saltwater is more dense than freshwater
    c. Groundwater extraction causes freshwater to rise
    d. Upwelling causes freshwater table to rise

35. Which of the following statements about aquifers are true?

I. The unsaturated zone contains no water, because it is above the water table.
II. Darcy's Law states that flow rate equals hydraulic conductivity times the driving force.
III. Unconfined and confined aquifers recharge in generally different methods.

    a. Only I is true
    b. Only II is true
    c. II and III are true
    d. All of the statements are true

36. Why is the water quality of karst-produced drinking water particularly vulnerable to contamination?
    a. Rocks like limestone and gypsum are prone to chemical contamination
    b. Sinkholes introduce vast amounts of pollutants into the water
    c. The high porosity of karsts allows water to flow quickly with little filtration
    d. Disappearing streams expose the water to foreign elements

37. What purpose do moraines primarily serve in the formation of glacial lakes?
    a. Moraines build up into natural dams, which transform streams into lakes
    b. Moraines absorb the meltwater from the glacier, creating an optimal environment for lake formation
    c. Moraines cause glaciers to partially melt, which leads to lake formation
    d. Moraines depress the land so that incoming water can be collected into a lake

38. Which of the following is a significant difference between barrier lakes and karst lakes?
    a. Barrier lakes are man-made, whereas karst lakes are natural
    b. Barrier lakes are formed by deposition, whereas karst lakes are formed by erosion
    c. Barrier lakes are formed when deposits block stream flow, whereas karst lakes are formed when moraines dam streams
    d. Barrier lakes are formed by tectonic movement, whereas karst lakes are formed by aeolian movement

39. Which of the following statements about lake stratification are true?
    I. Lake turnover can only occur when the entire body of water is the same temperature and density
    II. Lake turnover replenishes dissolved oxygen levels in deeper layers of the water
    III. The thermocline occurs within the epilimnion

    a. Only I is true
    b. I and II are true
    c. II and III are true
    d. All of the statements are true

40. Water flows from point A, which is 10 meters below a horizontal land surface, to point B, which is 15 meters below the same surface. The horizontal distance between point A and point B is 500 meters. The discharge rate of this water is known to be 10^{5} meters per second. Calculate the hydraulic conductivity of the aquifer.
    a. 10^{-3} m/s
    b. 2.5×10^{-3} m/s
    c. 10^{-4} m/s
    d. 2.5×10^{-4} m/s

41. Which of the following is most likely to cause stream piracy?
    a. Tectonic uplift, which causes damming of an existing stream and forces it to take another course
    b. Glacial damming, which causes damming of an existing stream and creates a glacial lake
    c. Sinkhole formation, which redirects a stream to a lower elevation
    d. Confluence with a Yazoo tributary

For questions 42-44, identify the type of instrument described.

42. This type of streamgage is operated only for given flow regimes such as high or peak flow.
    a. Seasonal streamgage
    b. Rapid-deployment gage
    c. Partial-record gage
    d. Supergage

43. This device constricts the width of the stream it is placed in, causing water level to change and allowing users to calculate streamflow through the device.
    a. Acoustic Doppler Current Profiler
    b. Weir
    c. Pygmy meter
    d. Flume

44. This device can be thought of as a tiny dam, with a spillway for water to discharge. Water height is measured at the spillway to calculate streamflow.
    a. Acoustic Doppler Current Profiler
    b. Weir
    c. Pygmy meter
    d. Flume

45. Where are dimictic lakes usually found?
    a. temperate regions
    b. polar regions
    c. tropical regions

46. Holomictic lakes have layers that mix, whereas meromictic lakes have layers that do not mix. What causes this difference?
    a. salinity
    b. temperature
    c. solar insolation
    d. dissolved nutrients

47. Density _____ with increasing salinity and ____ with increasing temperature.
    a. increases, increases
    b. increases, decreases
    c. decreases, increases
    d. decreases, decreases

48. When does the hypolimnion have the least dissolved oxygen?
    a. winter
    b. spring
    c. summer
    d. autumn

49. In the spring, lake turnover allows _____ from surface water to mix with the bottom layers and _____ from the bottom waters to mix with surface water.
    a. nutrients, dissolved oxygen
    b. dissolved oxygen, nutrients
    c. dissolved oxygen, sediment
    d. sediment, dissolved oxygen

50. Bogs are
    a. minerotrophic
    b. ombrotrophic
    c. amyotrophic
    d. hypertrophic

51. Marshes are
    a. minerotrophic
    b. ombrotrophic
    c. amyotrophic
    d. hypertrophic

52. Bogs are _____ while marshes are _____.
    a. nutrient rich, nutrient poor
    b. nutrient poor, nutrient rich

53. Down-cutting is the
    a. removal of forests downstream of a dam
    b. removal of vegetation near a riverbank
    c. removal of sediment in a stream bed

54. Which of the following is not a negative environmental effect of levees?
    a. increased erosion
    b. increased downstream flooding
    c. decreased groundwater recharge
    d. decreased dissolved oxygen

55. **(TB1)** Natural levees are formed by sediment deposits from repeated flooding. The sediment deposits near the riverbank because of
    a. higher friction between soil and water closer to the riverbank
    b. higher velocity of water closer to the riverbank
    c. decreased density of soil near the riverbank
    d. decreased vegetation near the riverbank

56. If evaporation is greater then precipitation,
    a. dissolved oxygen increases
    b. dissolved oxygen decreases
    c. salinity increases
    d. salinity decreases

57. A source of chloride in groundwater is
    a. rhizomes in legume roots
    b. gasoline leaks
    c. saltwater intrusion
    d. glass manufacturing

58. Increased vegetation _____ evapotranspiration, which _____ the infiltration rate.
    a. increases, increases
    b. increases, decreases
    c. decreases, increases
    d. decreases, decreases

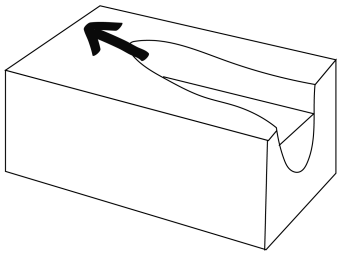
59. Soil permeability _____ after flooding
    a. increases
    b. decreases
    c. stays the same

60. Which of the following is a true statement about pingos?
    a. Pingos are commonly found in the subtropics
    b. Groundwater pumping causes subsidence, leading pingos to sink
    c. Climate change causes ice melt, leading pingos to collapse
    d. Pingo shape is determined by soil type, precipitation, and salinity.

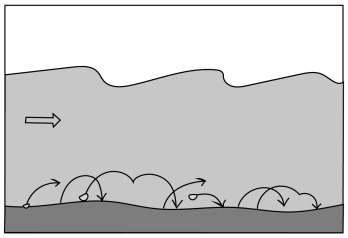
Matching (6 pts, 1 pt. for each correct answer)

Match each diagram with the process it best depicts. Write your answers on the provided answer sheet.

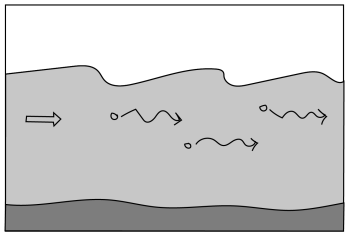
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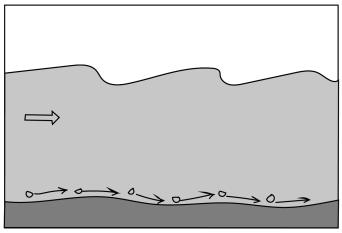
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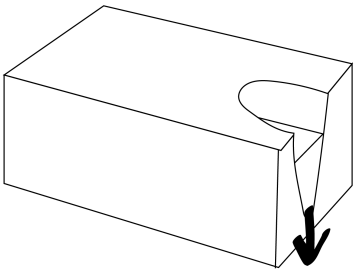
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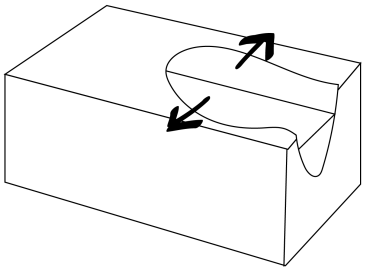
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65.



66.



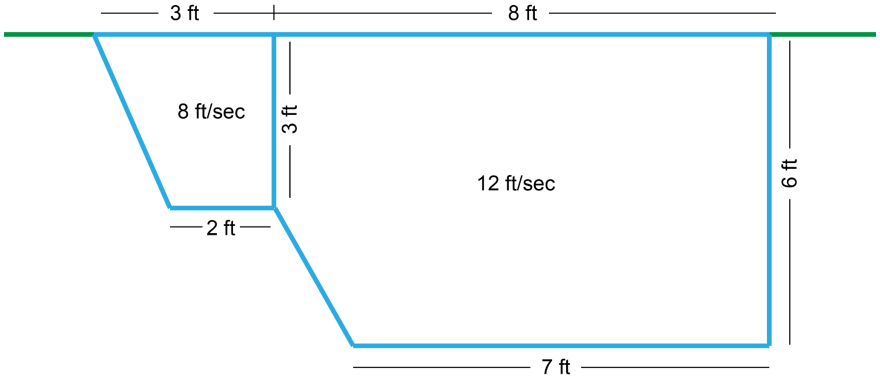
a. Suspension
b. Saltation
c. Headward erosion
d. Downcutting
e. Traction
f. Lateral Erosion

True/False (11 pts, 1 pt. for each correct answer)
Write your answers on the provided answer sheet.

67. Watersheds cannot exist within other watersheds.
68. The term "100-year flood" refers to a prolonged period of flooding caused by higher than average precipitation or water level rises.
69. Sediment can move between bed load and suspended load.
70. For streams that empty into an ocean, the base level is sea level.
71. Areas that have been damaged by wildfires are more susceptible to flash floods.
72. Chezy and Manning equations are typically used to calculate the discharge and velocity of groundwater moving from high to low elevations.
73. Large particles of sediment settle more slowly than small particles because their surface area is greater.
74. Weirs are used to calculate streamflow by collecting water volume data.
75. Competence is a constant number for any given stream.
76. Recurrence intervals should not be calculated with less than ten years of data.
77. An aquifer with high porosity will also have high permeability.

Short Answer (83 pts total, point value varies for each question)
Write your answers in the test booklet (this packet).

78. Calculate the stream discharge based on the following cross-section: (4 pts)

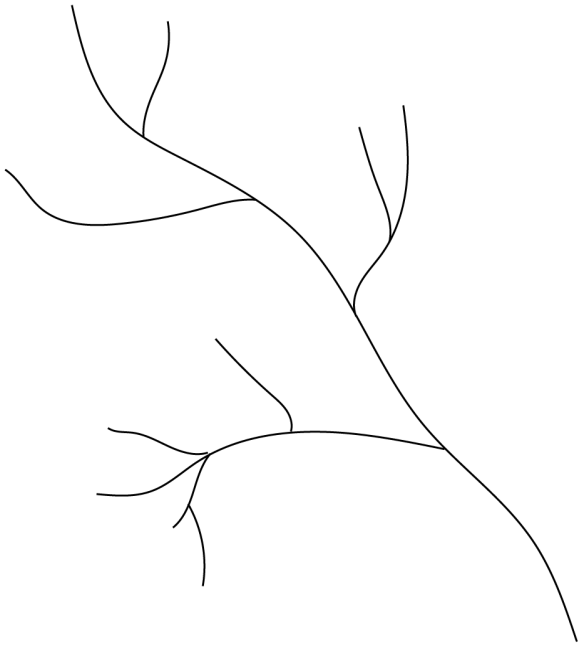


Discharge (with units) = _____

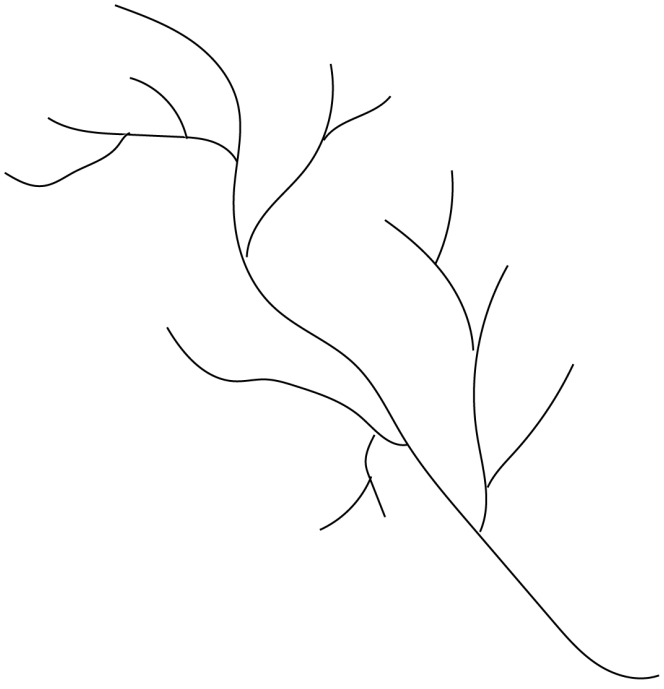
79. Describe the difference between suspended load, bed load, and dissolved load. (5 pts)

80. Number the stream orders for each branch of the streams in the given method. (6 pts)

a. Strahler



b. Shreve



81. Label each of the lettered areas with a term from the word box. (9 pts)

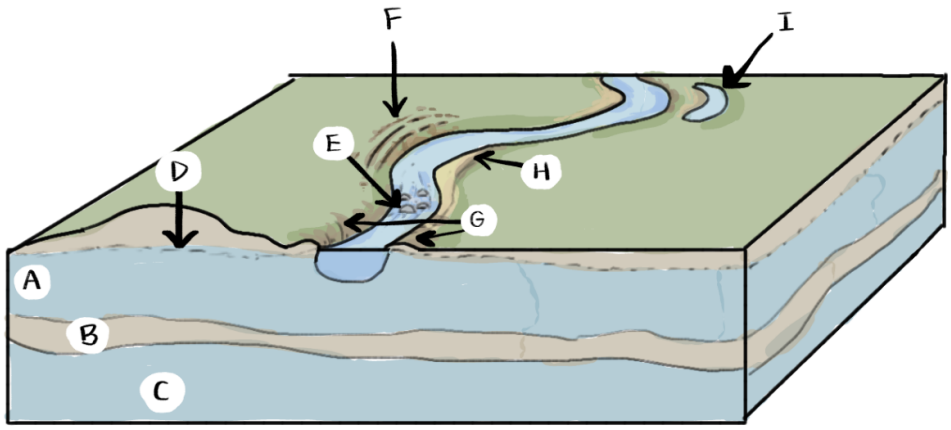
Word Box

unconfined aquifer	step pools	cutoff
confined aquifer	confining bed	riffle
alluvium	levees	pool
alluvial fan	watershed	anastomosing channel
water table	point bar	
terrace	beach	

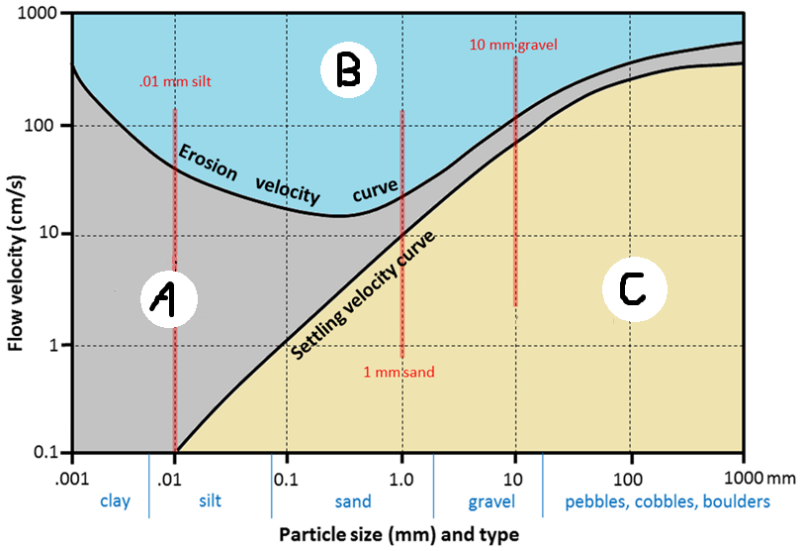
A: _____ D: _____ G: _____

B: _____ E: _____ H: _____

C: _____ F: _____ I: _____



82. (3 pts)



a. Identify the type of particles that represent each lettered region of the Hjulström-Sundborg diagram.

A: _____

B: _____

C: _____

b. Describe how the following particles would behave in a stream of flow velocity 100 cm/s.

i. Silt

_____

ii. 10mm gravel

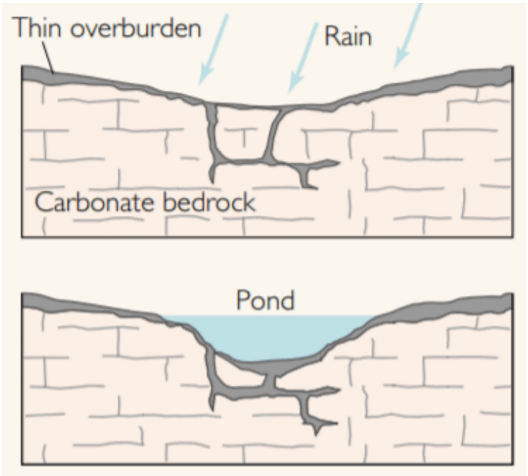
_____

iii. Pebbles, cobbles, and boulders

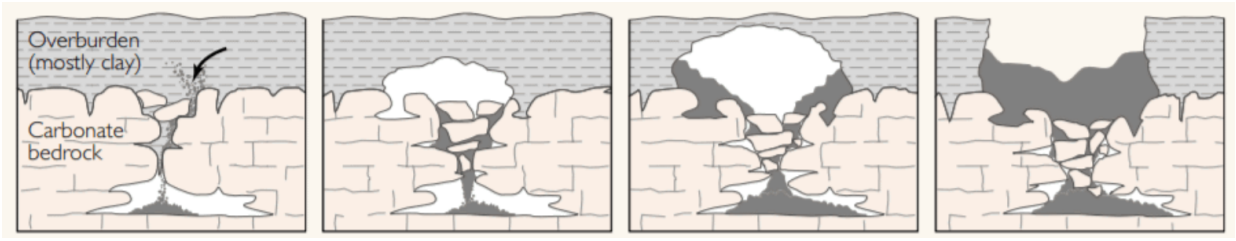
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83. Identify the type of sinkhole formation each diagram depicts. (3 pts)

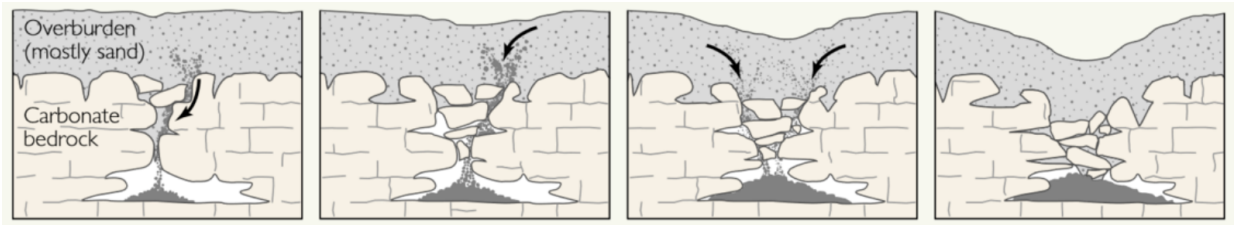
a.



b.



c.



84. What are the four inputs of water to a lake? (4 pts)

85. What is a dimictic lake? (4 pts)

86. In a mid-latitude lake, how is the thermocline different in the winter and summer? Why? (4 pts)

87. How does the addition of particulate matter to a lake affect its thermal stratification? (4 pts)

88. Describe how road salt runoff affects the vertical stratification of a lake. (4 pts)

89. What is the pH of bog water? What is the pH of marsh water? (No need for precise numbers, just acidic/basic/neutral is okay.) (2 pts)

90. Describe the two ways in which bogs can form. (4 pts)

91. Describe four negative environmental effects from the construction of dams. (2 pts)

92. For each of the following latitudinal regions, is precipitation or evaporation greater on average? (Can just say **P** if precipitation is greater and **E** if evaporation is greater.) (3 pts)
    a. polar

    b. subtropics

    c. equator

93. Write an equation representing the water budget of a watershed. (2 pts)

94. **(TB3)** How does the water budget balance if precipitation is greater than evaporation in a certain region (vertical column in atmosphere). What if evaporation is greater than precipitation? (4 pts)

95. Define point-source and nonpoint-source pollution. Give an example of each. (4 pts)

96. Eutrophication means an influx of nutrients that causes high biological productivity. Is this good or bad for a lake ecosystem? Why? (3 pts)

97. What is infiltration capacity? How does it change with soil moisture? (3 pts)

98. Name four factors that affect the rate of water infiltration through soil. (2 pts)

99. Describe the feedback loop caused by the melting of permafrost. Is this a positive or negative feedback loop? (4 pts)

Bonus Question: What is cabbeling? (brownie pts)